



EURO

Fundamental Quantum Computing Algorithms and Their Implementation in Qiskit

Michal Belina

IT4Innovations
VSB - Technical University of Ostrava

18. - 19.3. 2025



IT4INNOVATIONS
NATIONAL SUPERCOMPUTING
CENTER

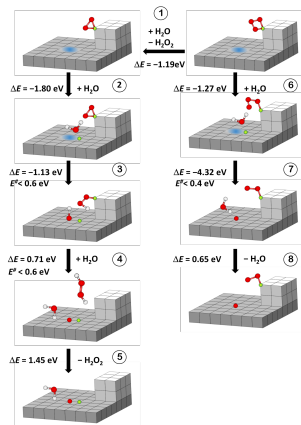
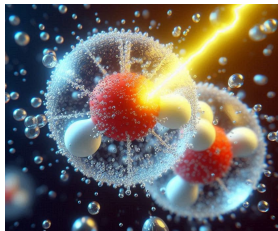


University of Chemistry and Technology Prague
(2014-2025):

- Physical chemistry
- Computational chemistry

VŠB-Technical University Ostrava (2024 - present):

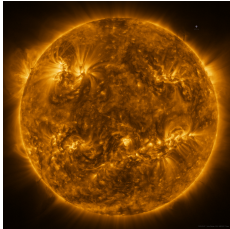
- Quantum Computing Lab



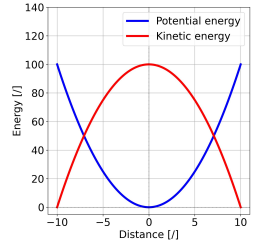
Classical vs Quantum Mechanics

Classical Mechanics

Sun



Chevrolet Impala 1967



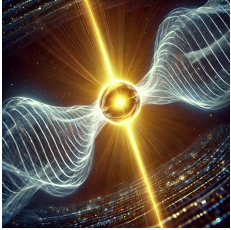
What would happen with the decreasing size?

- Weight: $\text{kg} \rightarrow 10^{-31} \times \text{kg}$
- Size: $\text{m} \rightarrow 10^{-10} \times \text{m}$

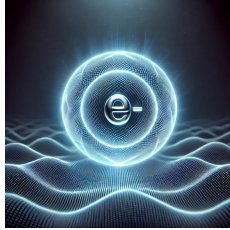
Classical vs Quantum Mechanics

Quantum Mechanics

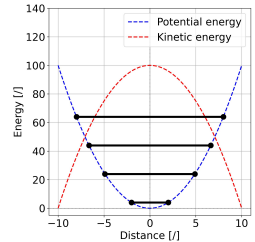
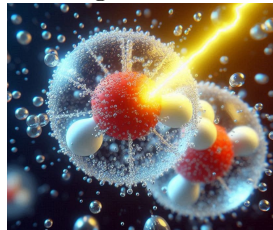
Photon



Electron



Light atoms

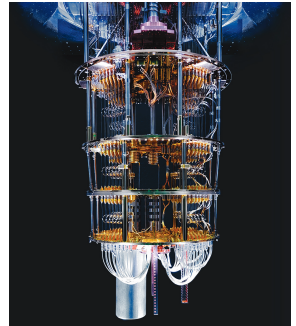
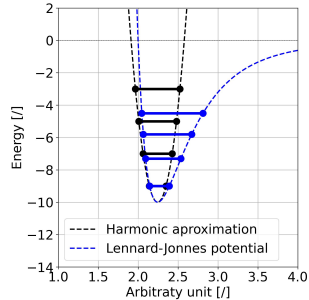
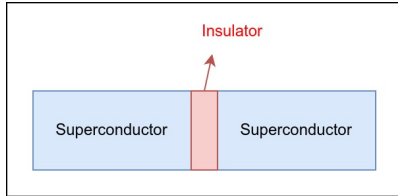


- Stern-Gerlach experiment
- Black-body radiation
- Zeeman effect

Quantum Hardware

Basic idea: To facilitate the discrete energetic levels

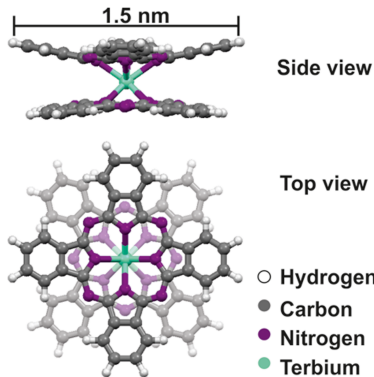
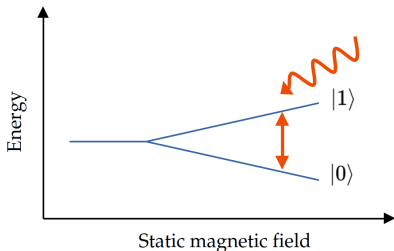
Superconducting technology - Josephson junction



Quantum Hardware

QC based on molecular spin

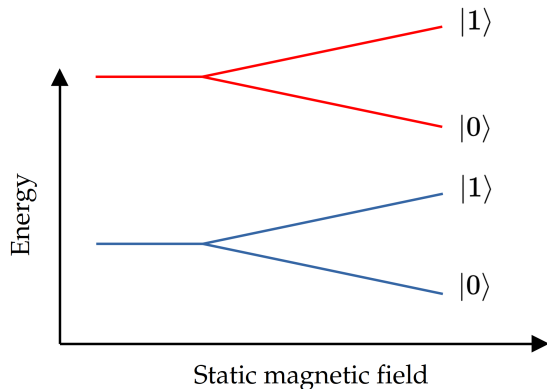
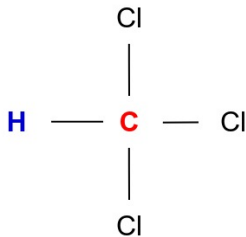
- Single-ion Magnets
- f-block ion in the middle – 1 “unpaired electron”
- Highly adjustable ligands



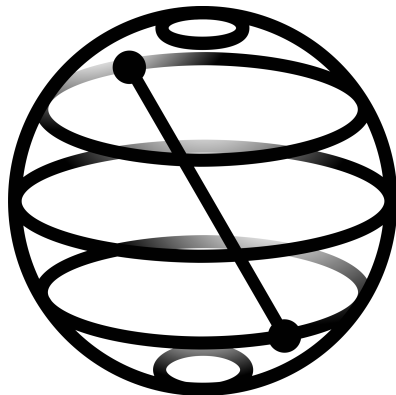
<https://pubs.acs.org/doi/pdf/10.1021/acs.jpcc.6b03676>

QC based on NMR

- Atoms with odd number of protons and/or neutrons.
- ^1H , ^{13}C ... ^{19}F , ^{31}P



- Qiskit = Quantum Information Software Kit
- open-source software development kit (SDK) for working with quantum Computers



Bloch sphere

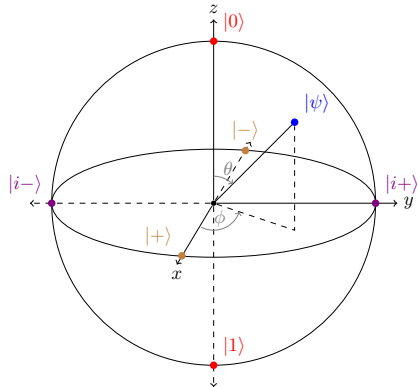
Other interesting point on Bloch sphere:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$|i+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

$$|i-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$



One-Qubit Quantum Gates

Definition: A one-qubit gate is represented by a 2×2 unitary matrix U acting on a single qubit:

$$U|\psi\rangle = U \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Important One-Qubit Gates:

- **Pauli Gates:**

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- **Hadamard Gate:**

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- **Phase Gates:**

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

Actions of One-Qubit Quantum Gates

Example:

- Pauli-X (NOT gate): $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$
- Hadamard on $|0\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

- Hadamard on $|1\rangle$:

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- Z-gate on superposition state:

$$Z \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Actions of One-Qubit Quantum Gates II

Initial State: The superposition state

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Action of the S Gate:

$$S|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |i+\rangle$$

Quantum Circuit:

$$|0\rangle \text{ --- } \boxed{H} \text{ --- } \boxed{S} \text{ --- } |i+\rangle$$

The S gate introduces a phase factor i to $|1\rangle$, transforming $|+\rangle$ into $|i+\rangle$, an eigenstate of the Y operator.

$$P(\lambda) |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\lambda} \beta \end{bmatrix}$$

$$Z |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

$$S |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix}$$

$$T |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\frac{\pi}{4}} \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \frac{1}{\sqrt{2}}(1+i)\beta \end{bmatrix}$$

Action of Gates on Special States:

$$Z |+\rangle = |-\rangle, \quad Z |-\rangle = |+\rangle, \quad S |+\rangle = |i+\rangle$$

$$Z |i-\rangle = SS |i-\rangle = TTTT |i-\rangle = |i+\rangle$$

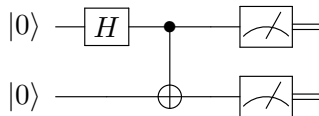
Quantum Circuits

Definition: A quantum circuit is a computational model in which quantum gates are applied to qubits to perform quantum computations.

Components of a Quantum Circuit:

- Qubits
- Quantum Gates (e.g., H , X , Z , $CNOT$, T)
- Measurements (collapsing qubits into classical bits)

Example: Bell State Circuit



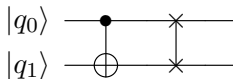
Common Two-Qubit Quantum Gates

- CNOT (Controlled-NOT) Gate:

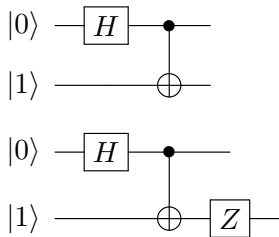
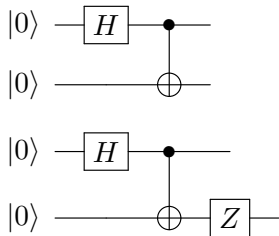
$$\text{CNOT} |11\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} |11\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle$$

- SWAP Gate:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Bell States



$$\begin{aligned} CX|H|00\rangle &= CX\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle\right) = \frac{1}{\sqrt{2}}CX|00\rangle + \frac{1}{\sqrt{2}}CX|01\rangle = \\ &= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |\Phi^+\rangle \end{aligned}$$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)$$